



Aircraft line maintenance scheduling and optimisation

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ABSTRACT

With the advent of new technologies and more modern aircraft, many of the maintenance jobs traditionally scheduled for periodic block checks can now be performed in the “line maintenance” environment, i.e., during layovers between scheduled flights of an aircraft. This flexibility can be exploited to reduce maintenance costs and improve fleet utilisation of an airline. In this paper we introduce and study the Line Maintenance Scheduling Problem (LMSP). The LMSP assigns jobs to available maintenance opportunities, defined by aircraft routes, and sets the starting time for each job. Its objective is to minimise the deviation from this schedule with respect to given due dates for each task, without exceeding resource capacity at the airports at any moment. We formulate the LMSP as a mixed integer programming problem, and describe and compare two solution approaches for this problem: an integrated exact solution algorithm, which solves job assignment and timetabling simultaneously, and a sequential, heuristic approach. We tested our algorithms on a set of instances inspired on data provided by an industry partner. Our experiments show the applicability of both approaches on realistic settings: the exact approach was able to find the optimal solution for all instances, in less than 10 min on average. Our analysis also shows with an example that line maintenance can be more efficient when capacity is spatially spread, even if the total capacity is reduced.

1. Introduction

Since the rise of commercial aviation in the years following World War II, safety in air travel has undergone an impressive evolution. For instance, the rate of yearly fatal accidents per million flights has dropped from around 10, in the sixties, to 0.05 in the present date (Airbus, 2020). This remarkable achievement would not have been reached without a collective effort and the implementation of very rigorous safety norms regarding aircraft maintenance.

Aircraft maintenance, as essential as it is, represents a big challenge for airlines in terms of costs and planning. Typically, aircraft maintenance costs make up about 10–20% of the costs associated with the operation of fleets of a typical airline (PeriyarSelvam et al., 2013). According to IATA (2019), global maintenance, repair and overhaul expenses in 2018 amounted to \$69 billion, with an annual growth rate of 4.1%. Hence, airlines are continually searching for ways to reduce this huge expense, while guaranteeing the airworthiness of their aircraft, and the safety of their passengers.

There are two main types of aircraft maintenance jobs: jobs that must be carried out in scheduled maintenance checks, and jobs that are

relatively simpler and can be performed routinely during the ground time of an aircraft. Maintenance jobs of the first type are bundled together in block checks, according to their periodicity and time requirements. For instance, an ‘A’ check is performed every two to three months, and takes the aircraft out of circulation for around six to 24 h, while on the other end of the spectrum ‘D’ checks are carried out every six to ten years, and require grounding an aircraft for several weeks (Deng et al., 2020).

“Line maintenance”, on the other hand, encompasses the execution of any minor maintenance activity on an aircraft right before a scheduled flight to ensure that it is airworthy for the intended flight operation. This may include defect rectification, troubleshooting, aircraft component replacement, general inspections, minor repairs and modifications. Aircraft line maintenance does not require extensive in-depth inspection, dis-assembly and rectification of aircraft and does not generally require hangar space unless weather and other special circumstances dictate. As more maintenance jobs now fall out of notional work blocks for modern aircraft, a better utilisation of the capacity in the line maintenance environment can help improve the efficiency of aircraft maintenance, and help balancing the workload at maintenance bases

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throughout the peak and trough periods (Hughes, 2006).

Although aircraft line maintenance generally requires neither an extensive investment of elapsed time or manpower for individual maintenance jobs, it contributes significantly to the maintenance cost due to its high frequency of occurrence with associated aircraft down time, skilled labour, materials and spare parts usage (Gupta et al., 2003). What makes line maintenance scheduling different from heavier maintenance scheduling is that the frequent job service requirements must be scheduled in the windows given by aircraft ground time intervals, which may overlap, and which compete for the resources available in the different airports where maintenance jobs are executed. This means that the job assignment to maintenance slots is not enough to determine if a schedule is feasible in terms of resource availability. The stochastic nature of ad hoc aircraft rectifications, the heterogeneity of aircraft fleets, and various cyclic maintenance constraints can make line maintenance scheduling even harder.

Line maintenance schedules are usually designed with the objective of maximising the maintenance yields while satisfying compliance deadlines and resource constraints. Technically, the term task yield refers to the ratio between the average interval of repetition versus the maximum allowed interval for a given task. So, if the objective of an airline was exclusively to save on maintenance costs, a myopic strategy would be to maximise yields and spread maintenance jobs apart in time as much as possible, i.e., to try to program jobs as close to their deadlines as possible. However, this approach would be extremely risky, because it leaves no room for recovering from last-minute disruptions. Although compliance extensions can be obtained in exceptional cases, they come with a compromised airworthiness situation, which may require special considerations, concessions and specific approvals from the airline's quality assurance department and the relevant civil aviation authority of the country. Non-compliance may involve contravention of the Civil Aviation Act or the relevant regulations of a country, and may involve a breach of the condition attached to an airline operating licence or certificate. Moreover, timely maintenance helps preventing aircraft systems to acquire a maintenance condition, similar to the way a human body can acquire a medical condition, which translates into higher maintenance costs in the long run.

To deal with these uncertainties airlines work with safety buffers, defined by a due time associated with each job. The ideal size of these buffers can be determined with the usual techniques used for defining safety stocks in the inventory management literature, and depends on the variability of the operation and the desired level of service to achieve. This way, the objective of the line maintenance problem can be reformulated as scheduling jobs as close to their due times as possible, which translates into minimising the total deviation from the "ideal" schedule defined by the due dates (or, the total earliness plus tardiness of the scheduled jobs).

Motivated by the importance and the challenging nature of line maintenance planning, we introduce a formal definition and mathematical formulation for the Line Maintenance Scheduling Problem (or just LMSP). Given a flight plan (or, directly, the set of available maintenance opportunities per aircraft), a fleet of aircraft, and a list of maintenance jobs to be carried out for each individual aircraft of this fleet, the objective of the LMSP is to determine when and where each maintenance job should be carried out within a planning horizon in order to minimise total deviation from due dates. The decisions involved in a solution of the LMSP can be classified in two levels: job assignment, and timetabling. On the top level, the job assignment stage decides which jobs are assigned to which available slot (or eventually postponed, if we allow jobs to be non-mandatory). In the timetabling phase, the objective is to set the specific starting times of jobs within their slots. Under this framework, the total deviation of a solution is determined by the job assignment, while the feasibility in terms of resource consumption and availability depends on the timetable, which ties the two decision stages together. Our formulation of the LMSP also considers that certain jobs can be postponed for execution beyond the planning

horizon.

To solve the LMSP, we propose a mixed integer programming (MIP) formulation of the problem, and devise and test two different solution algorithms. The first one tackles both levels of the problem in an integrated fashion, while the second one solves the problem sequentially: first it assigns jobs to available maintenance slots, and then it deals with the timetabling decisions. These two approaches pose a trade-off between the quality of the solution they reach (i.e., the total deviation from the ideal schedule) and the time they take to reach a solution. However, as we show later, for real-sized instances the integrated approach is able to find optimal solutions in about ten minutes of execution, which is very reasonable for a tool aimed at aiding a day-to-day decision making process.

This paper is organised as follows. §2 presents a literature review covering maintenance scheduling on different levels. In §3, a mixed integer linear programming formulation is proposed, which solves the complete problem to optimality. Then, §4 describes two solution approaches that are developed to solve the LMSP with short computation time. §5 presents the numerical experiments and benchmark policies used to compare the performance of two proposed solution methods. Results in terms of solution quality and execution times are discussed in §5 as well. Conclusions are then summarised in §6.

2. Literature review

The planning process of airline operations is a daunting task that in practice must be divided into smaller subproblems to make it tractable (Barnhart et al., 2003). The typical four stages of this planning process include flight scheduling, fleet assignment, crew scheduling, and aircraft maintenance and routing. This last stage deals simultaneously with the routing of specific aircraft, and the scheduling of mainly 'A' type checks. Line maintenance scheduling could be seen as a fifth stage in this planning hierarchy, dealing with shorter term decisions, when routes are already established, defining a set of maintenance opportunities or slots for each aircraft.

On a long-term planning perspective, where heavier or "structural" maintenance jobs are scheduled (i.e., checks type 'C' and 'D'), the literature is rather scarce. This happens because these decisions must be made several months or even years in advance, so they serve as an input to the longer term planning problems mentioned above. On an intermediate planning horizon, Deng et al. (2020) present a mathematical programming model that optimises the long-term integrated 'A' and 'C' checks schedule (they ignore 'B' checks, since they are infrequent in practice). Their model seeks to minimise the unused flight hours, which directly implies a reduction of the number of aircraft checks. The authors take into consideration the hangar capacity at each period, and include an inspection interval tolerance that allows scheduling checks after their deadlines to accommodate deviations from the initial schedule.

The literature in aircraft maintenance focuses mostly in the mid-term planning decisions, involving 'A' and 'B' block checks. Since hangar space is required to perform these routine maintenance checks, many authors have addressed this problem by integrating the decisions of aircraft routing and maintenance check scheduling in the so-called Aircraft Maintenance Routing Problem, or AMRP. This problem assigns a fleet of aircraft to cover scheduled flights, while fulfilling general maintenance constraints. One seminal work on the subject is that of Feo and Bard (1989), where the simultaneous problem of aircraft routing and maintenance planning is presented as a large-scale mixed integer linear programming model. Given a particular flight plan, this model assigns individual flights to each aircraft (i.e., aircraft routing), taking into account the costs associated with the optimal maintenance derived from that flight assignment. Since then, the AMRP has been studied by several authors, including Clarke et al. (1997); El Moudani and Mora-Camino (2000); Gopalan and Talluri (1998); Kabbani and Patty (1992) and Liang et al. (2012). All of these studies recognise that each

aircraft must visit a maintenance station periodically to perform the set of jobs defined by check 'A'. In the models they formulate, resource constraints are stated in an aggregated way (e.g., they may limit the number of aircraft that can undergo maintenance in a hangar during a day), without considering the individual processing times and resources that each job requires, or a detailed timetable for their execution.

The impact that unexpected disruptions cause in airline operations has raised the interest to introduce robustness in the maintenance planning process. Many authors have focused their work on the robust scheduling of 'A' and 'B' block checks. Lapp and Cohn (2012) present a model that maximises the likelihood of maintenance compliance by modifying the daily flight sequences (i.e., "line-of-flights") of each aircraft in case of disruptions. Liang, Feng, Zhang, Wu, and Chaovallitwongse (2015) propose a weekly line-of-flights network model to construct robust aircraft maintenance and routing schedules to minimise the total expected propagated delay. Yan and Kung (2016) consider the same objective function and explicitly model correlation in flight leg delays. Recently, Marla et al. (2018) study three different paradigms in the context of the robust aircraft maintenance and routing problem: domain-specific approach, an extreme-value based approach, and a chance-constrained or stochastic optimisation-based approach.

Even with robust planning, disruptions often force operators to adjust their plans to reschedule maintenance jobs. Sriram and Haghani (2003) combine the AMRP with an operational model that allows aircraft rerouting along with maintenance scheduling of checks 'A' and 'B'. Sarac, Batta, and Rump (2006) introduces the Operational Aircraft Routing Problem (or OAMRP). This problem consists in determining a valid daily aircraft routing considering the individual characteristics of each aircraft. The problem also recognises that some aircraft must finish their day-to-day operations at a maintenance station. The objective of this problem is to minimise unused flight hours before maintenance, subject to capacity constraints considered at an aggregate level. Haouari, Shao, and Sherali (2013) study the daily OAMRP where the schedule is repeated daily. The authors consider hangar capacity assuming that each 'A' type block has a fixed processing time. Başdere and Bilge, 2014 extend the work from Sarac et al. (2006) on a multi-commodity setting; Al-Thani et al. (2016) introduce a MIP formulation for the same problem and develop a very large-scale neighbourhood search (VLNS) heuristic to solve the problem.

Despite the importance of line maintenance as a medium for increasing efficiency in airline operations, the literature on this specific topic is scant. Senturk and Ozkol (2018) show that aircraft utilisation can be improved if individual jobs can be executed every time an aircraft is on the ground, compared to the traditional scheduling of periodic checks. This work, however, does not provide a mathematical formulation of the underlying problem. The few existing solution approaches for the LMSP found in the literature consist of simple heuristics. Papakostas, Papachatzakis, Xanthakis, Mourtzis, and Chrysosolouris (2010) present a decision-support tool that schedules maintenance jobs by focusing its analysis in one aircraft at a time, while fixing the rest of the schedule. Callewaert, Verhagen, and Curran (2018) develop real-time decision tool that integrates dynamic work progress monitoring data, and heuristically generates alternative plans when one or more immediate maintenance tasks deviate from their due dates. A dynamic take on the LMSP can be found in the work of Lagos et al. (2020). This work develops a model that selects, on a daily basis, a set of aircraft and jobs to be performed, aiming to minimise expected costs of expired maintenance jobs in the long term. To keep tractability, however, the problem is restricted to a single maintenance base.

Maintenance scheduling problems, and in particular the LMSP, are part of a vast family of scheduling problems, widely studied in the operations research literature. The LMSP is closely related to the Resource-Constrained Project Scheduling Problem (RCPSP). For a detailed review on the RCPSP and its extensions, please refer to Hartmann and Briskorn (2010). In a nutshell, the RCPSP aims to schedule a set of jobs, with known processing times and resource consumptions, assigning them to a

set of available machines. Jobs consume certain resources while being processed: the total consumption of each resource must not exceed its total capacity on the system. The LMSP can be understood as a variant of the RCPSP, where machines correspond to maintenance opportunities, jobs have no precedence constraints, and: i. Resource constraints apply to groups of machines (i.e., opportunities on the same airport); ii. Each job can only be assigned to a subset of specific machines (i.e., opportunities for the specific aircraft); iii. Machine availability is time-dependent (i.e., opportunities are available for a time window); iv. The objective is to minimise deviation from an ideal schedule. Some of these variants of the RCPSP have been studied in the literature. Drezeit and Billaut (2008) propose a MIP formulation and compare different solution heuristics for a time-dependent version of the problem with release dates and deadlines; Franck and Schwindt (1995) formulate a scheduling problem that minimises total earliness and tardiness of the schedule. As far as we could verify, there is no published material on a RCPSP with the characteristics of the LMSP.

3. Model development

In this section, we define and formulate the Line Maintenance Scheduling Problem (LMSP). This problem deals with two types of scheduling decisions: job assignments, and timetable design. In the job assignment problem, the LMSP assigns specific maintenance jobs to maintenance opportunities given by the aircraft schedules. In the timetabling problem, the model must define the time each job begins within its assigned slot. Each job has a due time when it should be ideally performed, so the objective of the job assignment is to minimise the total deviation from these due times. Additionally, each airport has a limited amount of certain resources required to perform each job: the timetabling decisions must guarantee that resource capacity is not exceeded at any time during the planning horizon. In what follows, we provide formal definition, and give a mathematical exact formulation for the LMSP.

3.1. Problem definition

Let us consider an airline operating with a certain fleet of aircraft, represented by set F . Each aircraft $f \in F$ has a set of maintenance jobs J_f to be carried out during a planning horizon of length L . The full set of jobs for the whole fleet is denoted by $J = \bigcup_{f \in F} J_f$. Each job has a duration of T_i . Jobs can be divided in two different groups, namely mandatory jobs O , and non-mandatory jobs N .

For every mandatory job there is a known release time Q_i , a due time C_i , and a deadline D_i . A mandatory job must be performed after its release time and before its deadline, and ideally as close as possible to its due time. Due to the diverse nature of maintenance jobs, these three values (Q_i , C_i and D_i) can be expressed in different units. Each job has thus associated its own relevant unit. In this work we consider jobs with three possible relevant units: flying hours, number of take-off cycles, or calendar days. Note that this distinction is necessary because these values increase in different rates, depending on the flight itineraries. For example, when an aircraft is on the ground, its flight time remains constant, but calendar days keep increasing.

Non-mandatory jobs, on the other hand, have no release time, due time or deadline. These jobs can be left unscheduled at a cost of c_i . This cost penalty could represent the cost of hiring additional staff to carry out the job, or outsourcing this job to a third party company.

Every job consumes a certain amount of different types of resources during its maintenance, such as specific types of trained personnel, equipment or materials. We define the set K as the set of resources that jobs may require. We also define μ_i^k as the amount of resource k required by job i at any time during its maintenance.

A job $i \in J_f$ can only be executed when aircraft f is on the ground at an airport. We denote the set of airports or maintenance bases as P . The

complete set of maintenance opportunities for the whole fleet is denoted as M . We also define M_i as the subset of M where job i can be performed. The starting and finishing times of a maintenance opportunity are denoted by l_m and u_m .

We define parameter t_{im} as the utilisation of the aircraft on which mandatory job i must be performed, at the beginning of opportunity $m \in M_i$. This parameter, which can be obtained from the flight schedule of the aircraft, is expressed in the relevant unit for the corresponding job, and is used for measuring deviation from the due time. For every mandatory job, parameters a_i and b_i indicate the cost of earliness or tardiness respective to the due time, weighted in order to convert their relevant units in a common measuring unit.

A feasible solution for the problem must not exceed resources available at the airports where the jobs take place. Set M_{ip} contains the maintenance opportunities available for job i at a given airport p . Airport p has a supply of R_p^k units of resource $k \in K$ at any moment during the planning horizon. This value cannot be exceeded by the total consumption of resources of the different jobs being performed at any given time at the airport. Note that we assume in this formulation of the LMSP that resource availability is not time-dependent, to simplify the formulation and solution of the problem.

To deal with non-mandatory jobs, we define for $i \in N$ a fictitious maintenance opportunity m_i^0 . This maintenance opportunity takes place in a fictitious airport denoted as $p = 0$, starting at time $l_{m_i^0} = 0$ and ending at time $u_{m_i^0} = T_i$. Fictitious airport $p = 0$ has no resource limitations, so that any non-mandatory job can be assigned to its fictitious maintenance opportunity without generating any conflicts.

A solution for the LMSP is a maintenance schedule determined by three decisions: whether or not to perform each non-mandatory job; the maintenance opportunity where jobs are scheduled (if they are scheduled); and the exact time the jobs are scheduled to begin (if they are scheduled). The objective of the problem is to find the maintenance schedule that minimises the total deviation of mandatory jobs, plus the penalty for delayed non-mandatory jobs, without exceeding resource capacity at the airports.

It is worth noting that in this version of the LMSP model, if resource constraints are non binding, all jobs are independent from each other. This leaves out of the current problem coordination issues like precedence relations between jobs, or even interdependence between deadlines of related jobs of the same aircraft (i.e., rushing a job may bring future deadlines of the same job earlier).

3.2. Mathematical formulation of the LMSP

Due to the varied nature of the problem (job assignment, time windows and resource constraints), we propose a mixed integer programming (MIP) formulation for the LMSP. The sets and parameters involved in this MIP formulation are summarised below.

Sets:

- O: Mandatory jobs.
- N: Non-mandatory jobs.
- J: Set of all jobs (i.e., $J = O \cup N$).
- P: Set of airports (or maintenance bases), not including fictitious airport $p = 0$.
- F: Set of aircraft.
- M: Maintenance opportunities.
- M_i : Maintenance opportunities for job i , including a fictitious one when $i \in N$.
- M_{ip} : Maintenance opportunities for job i taking place in airport p .
- K: Types of resources consumed during maintenance jobs.

Parameters:

- a_i : Cost per unit of earliness for job i .

- b_i : Cost per unit of tardiness for job i .
- c_i : Cost for postponing a non-mandatory job i .
- D_i : Deadline of job i .
- C_i : Due time of job i .
- Q_i : Release time of job i .
- T_i : Processing time of job i .
- r_i^k : Quantity of resource k consumed by job i .
- R_p^k : Quantity of resource k available at airport p .
- l_m : Starting time of maintenance opportunity m .
- u_m : Finishing time of maintenance opportunity m .
- t_{im} : Utilisation of the aircraft on which mandatory job i must be performed, when arriving at maintenance opportunity m .
- L: Duration of the planning horizon.
- ϵ : Tolerance value used in strict inequalities.

The variables used in the LMSP are listed below. The first two variables we introduce represent the two main decisions that determine a maintenance schedule: where to assign jobs (with the possibility of postponing them), and when to start their execution. The list of variables then continues with some auxiliary ones, included to simplify the formulation of the model.

Variables:

- x_{im} : binary, indicates if job i is executed during maintenance opportunity $m \in M_i$.
- s_i : instant when job i starts.
- e_i : instant when job i finishes.
- y_i : earliness of job i .
- z_i : tardiness of job i .
- w_i^p : binary, indicates if job i is executed at airport p .
- A_{ij} : binary, indicates if job j begins before job i starts, i.e., if $s_j \leq s_i$.
- B_{ij} : binary, indicates if job j finishes after job i starts, i.e., if $s_i < e_j$.
- G_{ij} : binary, indicates if job j is being executed when job i starts, i.e., if $s_j \leq s_i < e_j$.
- F_{ij}^p : binary, indicates if jobs i and j are both carried out in airport p .
- H_{ij}^p : binary, indicates if job j is being executed when job i starts and both jobs are carried out in airport p .

In this list, the group of binary variables notated in upper cases serve a common purpose: to determine which jobs are taking place at an airport at the moment a given job is scheduled to start there. The idea behind this is that, to ensure that the resource constraint holds during the whole planning horizon, it is enough to verify it holds at the beginning of every job taking place in the same airport. Note also that the inequality that defines A_{ij} is not strict, while the one defining B_{ij} is. This happens because if a job ends at the same time another one begins, they will not compete for resources.

3.3. Mathematical formulation

We now formulate the LMSP as a MIP problem. The objective function to minimise in the LMSP is the sum of earliness and tardiness of mandatory jobs, which defines the total deviation of a solution, plus the cost of non-mandatory jobs left unscheduled:

$$\min \sum_{i \in O} (a_i y_i + b_i z_i) + \sum_{i \in N} c_i x_{im_i^0} \quad (1)$$

Note that variable $x_{im_i^0}$ in this expression indicates if a non-mandatory job i is assigned to its fictitious maintenance opportunity m_i^0 , i.e., if job i is left unscheduled. A feasible solution for the problem is defined by the following set of constraints:

$$\sum_{m \in M_i} x_{im} = 1, \quad \forall i \in J, \quad (2)$$

$$z_i - y_i = \sum_{m \in M_i} (x_{im} t_{im} - C_i), \quad \forall i \in O, \quad (3)$$

$$\sum_{m \in M_i} x_{im} t_{im} \geq Q_i, \quad \forall i \in O, \quad (4)$$

$$x_{im} t_{im} \leq D_i, \quad \forall i \in O, m \in M_i, \quad (5)$$

$$s_i + T_i = e_i, \quad \forall i \in J, \quad (6)$$

$$\sum_{m \in M_i} x_{im} l_m \leq S_i, \quad \forall i \in J, \quad (7)$$

$$e_i \leq \sum_{m \in M_i} x_{im} u_m, \quad \forall i \in J, \quad (8)$$

$$\sum_{m \in M_{ip}} x_{im} = w_i^p, \quad \forall i \in J, p \in P, \quad (9)$$

$$\sum_{j \in J, j \neq i} H_{ij}^p r_j^k + w_i^p r_i^k \leq R^{kp}, \quad \forall i \in J, k \in K, p \in P, \quad (10)$$

$$s_i - s_j \leq L \cdot A_{ij} - \varepsilon, \quad \forall i \in J, j \in J, j \neq i \quad (11)$$

$$e_j - s_i \leq L \cdot B_{ij}, \quad \forall i \in J, j \in J, j \neq i \quad (12)$$

$$G_{ij} \geq A_{ij} + B_{ij} - 1, \quad \forall i \in J, j \in J, j \neq i \quad (13)$$

$$F_{ij}^p \geq w_i^p + w_j^p - 1, \quad \forall i \in J, j \in J, j \neq i, p \in P, \quad (14)$$

$$H_{ij}^p \geq F_{ij}^p + G_{ij} - 1, \quad \forall i \in J, j \in J, j \neq i, p \in P, \quad (15)$$

$$s_i, e_i, y_i, z_i \geq 0, \quad \forall i \in J, \quad (16)$$

$$w_i^p, x_{im}, A_{ij}, B_{ij}, G_{ij}, F_{ij}^p, H_{ij}^p \in \{0, 1\}, \quad \forall i \in J, j \in J, p \in P, m \in M. \quad (17)$$

The first four constraints in the model deal with the assignment of jobs to maintenance opportunities. Equation (2) forces every job to be assigned to a maintenance opportunity (possibly a fictitious one). Expression (3) computes the earliness or tardiness of mandatory maintenance jobs. Constraints (4) and (5) ensure that mandatory jobs are executed some time between their release time and their deadline.

Constraints (6) to (15) deal with the timetabling decisions, i.e., the precise time at which jobs take place within their maintenance opportunities. Constraints (6) link together the start and finish time of jobs. Constraints (7) and (8) ensure that jobs begin and end within the limits of the maintenance opportunity where they are assigned. Expression (9) defines the airport where a job is executed.

Constraints (10) to (15) ensure the feasibility of the operational schedule in terms of resource consumption at any airport, at any moment of the planning horizon. Expression (10) checks that resource consumption does not exceed capacity every time a new job i starts in an airport. This value is computed as the sum of resources consumed by job i , plus the resource consumption of jobs j that overlap with i in that particular moment. The set of jobs that overlap with i is represented by the auxiliary variable H_{ij}^p , which is constructed by expressions (11) to (15). First, constraints (11) to (13) build auxiliary variable G_{ij} which indicates if job j is being executed at the moment job i starts. The use of the tolerance value ε in expression (11) forces A_{ij} to take value one even when j starts at the same time as i . Then, expression (14) builds auxiliary variable F_{ij}^p that indicates if both jobs are executed at the same airport p . Finally, expression (15) sets variable H_{ij}^p to one if job j is being executed when job i starts (i.e., if $G_{ij} = 1$) and both jobs take place in the same airport (i.e., $F_{ij}^p = 1$). The remaining constraints (16) and (17) establish the non-negative and/or binary nature of each variable.

The MIP formulation of the LMSP is thus defined as the minimisation

of the objective function (1) subject to constraints (2) to (17). In the following section we discuss different solution approaches for this problem.

4. Solution approaches

In this section we provide two different solution approaches for the LMSP. Given that the problem combines two identifiable decision levels, our approaches study and exploit the advantages that may come from separating or combining these two stages. The first approach we report is an exact method that solves both levels of the problem simultaneously by solving the MIP model described in the last section. The second approach is a greedy-type heuristic that solves the problem sequentially, focusing first on assigning jobs to maintenance opportunities, and then on the timetabling decisions. This second approach resembles closely the current practice from our partner in the industry, which means it can be used as a benchmark for the exact solution of the integrated problem.

4.1. Exact solution algorithm for the integrated LMSP (ILMSP)

The first solution approach consists, as mentioned above, in solving the MIP model given by minimising (1) subject to constraints (2) to (17). This MIP formulation tackles both planning levels of the problem (i.e., job assignment and timetabling) in an integrated way. Nonetheless, in order to exploit this layered structure of the problem, we designed an exact solution algorithm that focuses initially on job assignment, and then dynamically incorporates violated operational scheduling constraints to find the integrated optimal solution.

The solution algorithm for the integrated LMSP (or ILMSP) starts by dropping constraints (10) to (15) from the original LMSP. Relaxing these constraints, which are the ones related to the operational scheduling dimension of the model, reduces the LMSP to an assignment problem. Then, this relaxed LMSP is solved using a branch-and-bound algorithm, using a callback function to find and reincorporate violated capacity constraints (10) and their corresponding auxiliary variable constraints (11) to (15).

More precisely, whenever we find a solution to the assignment problem (which might already include some capacity constraints), we check if this solution is feasible for the original ILMSP. To do so, we compute the resource requirements for the current solution. We do this for each airport, for each type of resource, whenever a job is scheduled to start in the solution. These requirements are contrasted against the resource availability, to determine if the solution is feasible and, if it is not, generate a group of capacity constraints to reincorporate, to prevent the conflict in the current solution from happening.

For instance, if we detect that capacity is exceeded in airport $\hat{p}$ for resource $\hat{k}$ when job $\hat{i}$ is set to start, we proceed as follows:

- Check which other jobs generate a conflict with job $\hat{i}$ in the solution, i.e., which other jobs are consuming resource $\hat{k}$ at airport $\hat{p}$ at the time job $\hat{i}$ begins. We will call this the set of conflicting jobs, $\mathcal{C}$.
- Generate and incorporate constraints (10) to (15), for the conflicting jobs, in airport $\hat{p}$ for resource $\hat{k}$:

- Constraint 10: $\forall i \in \mathcal{C}, k = \hat{k}, p = \hat{p}$.
- Constraints 11, 12, and 13: $\forall i \in \mathcal{C}, j \in J, j \neq i$.
- Constraints 14 and 15: $\forall i \in \mathcal{C}, j \in J, j \neq i, p = \hat{p}$.

4.2. Heuristic solution for the sequential LMSP (SLMSP)

The SLMSP heuristic was developed to mimic the current practice of maintenance schedulers of the study airline that provided a data set for research purposes. This practice consist in manual procedure resembling a greedy algorithm. The procedure starts with allocating jobs with

deadlines and with high priorities first, then move on to jobs with soft deadlines and/or lower priorities, and then to jobs without deadlines and low priorities. This phase of job allocation ensures that maintenance compliance is met as much as possible with highly prioritised jobs being allocated first. At this stage the resource capacity of each maintenance opportunity is roughly estimated, in order to produce solutions that are more likely to be feasible. This first phase produces an initial solution for the line maintenance problem. Then, the manual procedure starts a second phase of ad-hoc schedule tuning, in which maintenance planners fine-tune the solution from the first phase to guarantee resource availability, according to in-house scheduling preferences such as varying the due-time buffer or manually reallocating jobs to accommodate unscheduled jobs. Note that this manual procedure is not guaranteed to always find a feasible solution.

Based on this current practice, we developed a sequential heuristic to solve the aircraft line maintenance scheduling problem (the SLMSP algorithm). The aim is to develop SLMSP in such a way that it could be used as a decision-making support tool by the study airline to assist maintenance schedulers in their daily jobs. Since this algorithm mimics the current practice of schedulers, it makes it easier for schedulers to comprehend. When schedulers need to adjust the results produced by the algorithm, it is also easier for schedulers to understand the rationale as why a solution is generated in the first place.

The SLMSP algorithm can be thus summarised in two main stages, that deal sequentially with the job assignment and timetabling decisions of the LMSP:

1. Job assignment stage. In this stage, mandatory jobs are prioritised over non-mandatory ones, and ranked increasingly according to their deadlines. Then, jobs are assigned to maintenance opportunities in a greedy fashion, aiming to minimise their deviation from their due times. In this stage, capacity constraints are incorporated in an aggregated way, by comparing the total capacity of a maintenance slot (i.e., its level of resources times the duration of the opportunity) against the total demand of a job (i.e., its resource consumption times its duration).
2. Timetabling stage. This stage of the SLMSP heuristic takes the assignment of jobs to maintenance opportunities, and searches for a feasible timetable in terms of capacity consumption at every instant, at every airport or maintenance base, considering that jobs may occur simultaneously. This stage begins by setting all jobs to start at the beginning of their current maintenance opportunity. Then the heuristic looks for resource violations, i.e., instants of time where the consumption of a specific resource at a particular airport exceeds the available capacity. From the set of conflicting jobs generating this infeasibility, the algorithm selects the smallest one (in terms of resource consumption) whose removal would restate feasibility of the detected violation. The heuristic then attempts to reschedule the removed job, first within its current maintenance opportunity at different starting times, and then in its closest maintenance opportunity.

A more in depth description of the SLMSP algorithm is provided in [Appendix 1](#). The detailed version of this heuristic includes more sophisticated prioritisation criteria, aimed at capturing aspects of the real process followed by our partner airline. This algorithm, just like the procedure followed by the airline in practice, does not guarantee finding a feasible solution.

5. Numerical experiments

In this section, we present our numerical experiments, designed for proving the applicability of our solution algorithms in a planning operation. We use our algorithms to answer to two main research questions. The first one is to determine the benefits of solving the integrated scheduling problem (i.e., that combines job assignment with

timetabling) versus a benchmark sequential heuristic approach, inspired in the current practice of our partner in the industry. The second research question is to measure how the level of congestion of the system (i.e., the level of demand versus resources available) affects its performance.

5.1. Research data and test environment

To conduct our experiments, we built a set of randomly generated instances, replicating the structure of a set of real data provided by our partner airline. The available data consisted on detailed information regarding flight schedules and required maintenance jobs for a fleet of three aircraft. The airline operates with four maintenance bases, each with a given number of person-hours available for maintenance on three trade categories (airframe, engine and avionics). Job deadlines, due dates and release dates were measured in three possible relevant units, namely, flying hours, number of landing cycles, and calendar days. For practical reasons we only considered maintenance windows with a ground time of at least two hours.

The characteristics of each instance are summarised in [Table 1](#). The first column lists the instances and indicates the number of aircraft (A), the duration of the planning horizon in weeks (L), and the total number of jobs (J). Column two indicates the total number of jobs with deadlines in terms of cumulative flying hours ($|O_1|$), aircraft landing cycles ($|O_2|$) and calendar days ($|O_3|$), while column three shows the number of non-mandatory jobs $|N|$. Column four lists the number of maintenance opportunities $|M|$. The fifth column gives the average job processing time $\bar{T}$ in hours and the last column indicates the average amount of resources $\bar{r}^k$ (in the unit of person-hours) needed to execute a job in terms of the three maintenance categories: airframe (AF), engine (E) and avionics (Av), so the resource set is $K = AF, E, Av$. The set of instances covers a wide range of values in terms of number of jobs, number of aircraft, maintenance opportunities, job process times and resource requirements.

For all the instances, we consider that all maintenance bases have the

Table 1
Numerical test instances.

Instance	$(O_1 / O_2 / O_3)$	$ N $	$ M $	$\bar{T}$	$\bar{r}^k$
(A/L/J)				(hours)	(AF/E/Av)
(3/4/112)	(51/25/10)	26	160	1.53	(0.84/0.26/0.16)
(3/3/82)	(42/9/10)	21	138	1.84	(0.62/0.42/0.17)
(3/2/57)	(25/10/9)	13	79	1.68	(0.88/0.32/0.16)
(4/4/157)	(75/21/20)	41	209	1.77	(0.74/0.33/0.14)
(4/4/148)	(73/19/21)	35	223	1.58	(0.79/0.32/0.11)
(4/3/112)	(53/13/19)	27	155	1.96	(0.67/0.34/0.15)
(4/3/157)	(83/10/25)	39	165	1.8	(0.75/0.29/0.18)
(4/2/77)	(33/15/11)	18	116	1.41	(0.80/0.29/0.14)
(5/4/189)	(92/24/25)	48	253	2.03	(0.70/0.25/0.16)
(5/4/202)	(95/27/30)	50	283	1.54	(0.76/0.33/0.14)
(5/3/150)	(83/20/13)	34	210	1.69	(0.96/0.33/0.08)
(5/3/187)	(91/19/28)	49	207	1.58	(0.75/0.31/0.14)
(5/2/94)	(43/13/14)	24	131	1.8	(0.87/0.21/0.11)
(6/4/221)	(108/22/31)	60	352	1.61	(0.82/0.28/0.16)
(6/3/229)	(108/31/29)	61	242	1.59	(0.69/0.31/0.16)
(7/4/263)	(134/28/32)	69	399	1.78	(0.74/0.35/0.15)
(7/3/243)	(113/43/26)	61	292	1.8	(0.64/0.31/0.17)
(8/4/298)	(142/41/39)	76	451	1.71	(0.70/0.31/0.19)
(8/3/285)	(154/36/23)	72	319	1.61	(0.77/0.31/0.14)
(9/4/329)	(159/40/48)	82	499	1.79	(0.78/0.35/0.14)
(9/3/348)	(171/43/47)	87	365	1.82	(0.79/0.26/0.18)
(10/4/372)	(184/38/51)	99	527	1.78	(0.70/0.31/0.16)
(10/3/382)	(194/50/42)	96	414	1.78	(0.74/0.32/0.18)
(11/4/424)	(208/55/52)	109	612	1.42	(0.67/0.32/0.17)
(11/3/415)	(207/56/48)	104	464	1.7	(0.74/0.35/0.14)
(12/4/337)	(168/41/41)	87	646	1.71	(0.72/0.31/0.14)
(12/3/211)	(111/21/25)	54	509	1.92	(0.83/0.27/0.15)
(13/4/467)	(220/62/66)	119	722	1.82	(0.74/0.33/0.13)
(13/3/370)	(181/55/42)	92	545	1.66	(0.83/0.32/0.15)

same resource capacity: three person-hours for airframe, two for engine, and one for avionics (i.e., $R_p^{AF} = 3, R_p^E = 2, R_p^{Av} = 1, \forall p \in P$). We also consider that the earliness and tardiness cost is similar for any given job, i.e., $a_i = b_i, \forall i \in O$. To express our objective function in a standard unit, we use different weighting factors depending on the deadline type (flying hours, cycle, or days). From our data, we observe that during one day, the average aircraft utilisation was roughly 10 h, and each cycle takes approximately 2.5 h. Thus, we use the weighting factor 1, 2.5 and 10 to express earliness and tardiness in terms of hours, cycles and days, respectively. We also set $c_i = 50$ for all non-mandatory jobs $i \in N$ in the objective function, to encourage the scheduling of all jobs during the planning horizon. Due and release times were generated discounting a fixed amount from the generated deadlines. Due times were set to 15 h, 10 cycles or 1 day before the deadline, and release times were set to 50 h, 25 cycles or 5 days before the deadline, according to the relevant unit for the job.

5.2. Computational results

In this section we report the results of the experiments conducted on the set of instances presented in the previous section, using the exact, integrated algorithm (ILMSP), and the heuristic, sequential algorithm (SLMSP). The ILMSP was implemented in Python (v.2.7.10) and solved with Gurobi (v.8.1). The SLMSP was implemented on Microsoft Excel 2013 and solved with Microsoft Visual Basic. We imposed a maximum runtime of 3600 s (60 min) for all test instances. All experiments were performed on a computer equipped with an Intel Core i5 2.6 GHz processor and 8 GB of RAM.

5.2.1. Integrated versus sequential solution procedures

In this section we report and compare the results obtained from solving the LMSP in a set of instances using the integrated and sequential approaches (i.e., the ILMSP and SLMSP algorithms).

Table 2 summarizes the comparative results for the SLMSP and

Table 2
Comparative computational results for the SLMSP and ILMSP algorithms.

Instance	SLMSP			ILMSP ^a	
	O.F.	gap	Time (s)	O.F.	Time (s)
(A/L/J)					
(3/4/112)	148.0	0.0%	5	148.0	5
(3/3/82)	98.0	6.5%	4	92.0	13
(3/2/57)	66.0	0.0%	3	66.0	6
(4/4/157)	204.5	0.0%	7	204.5	10
(4/3/112)	157.0	4.7%	4	150.0	8
(4/2/77)	58.0	0.0%	3	58.0	14
(5/4/189)	307.0	0.5%	6	305.5	42
(5/3/150)	197.0	2.1%	5	193.0	10
(5/2/94)	134.5	8.0%	2	124.5	5
(4/4/148)	155.0	0.0%	7	155.0	10
(4/3/157)	214.5	1.4%	7	211.5	91
(5/4/202)	195.5	0.5%	10	194.5	92
(5/3/187)	235.5	5.8%	9	222.5	90
(6/4/221)	286.0	0.0%	11	286.0	400
(6/3/229)	254.0	2.8%	11	247.0	209
(7/4/263)	303.5	1.5%	14	299.0	892
(7/3/243)	309.5	0.0%	11	309.5	236
(8/4/298)	385.0	4.3%	15	369.0	124
(8/3/285)	362.0	1.4%	12	357.0	685
(9/4/329)	428.5	10.4%	16	388.0	938
(9/3/348)	477.0	2.8%	16	464.0	1230
(10/4/372)	516.0	7.1%	17	482.0	781
(10/3/382)	520.0	6.0%	18	490.5	3600
(11/4/424)	447.5	3.0%	20	434.5	1223
(11/3/415)	681.5	11.5%	18	611.0	1791
(12/4/337)	475.5	5.9%	13	449.0	329
(12/3/211)	242.5	1.9%	10	238.0	123
(13/4/467)	554.5	4.2%	22	532.0	1822
(13/3/370)	509.5	8.6%	16	469.0	743
Average	307.7	3.5%	10.8	294.9	535.2

^a The ILMSP algorithm solved every instance to optimality.

ILMSP algorithms. The first column lists the instances. Columns 2–4 provide the resulting value of the objective function (O.F.), the optimality gap of the solution, and the solution time obtained for the SLMSP. Columns 5 and 6 report the objective function value and solution time for the ILMSP algorithm. Every single solution reported of the ILMSP is globally optimal, which is why we omit the optimality gap for this algorithm (also, this objective function value was used to compute the gap from the SLMSP).

The first result that stands out from Table 2 is that the ILMSP clearly dominates SLMSP in terms of the quality of the solution. Not only the integrated approach reaches the global optimal solution on every single instance, its objective function value is on average 4.2% lower than the one obtained with the benchmark sequential approach. Nevertheless, the SLMSP is still able to reach optimality in seven out of 29 instances, with an average gap of 3.5%, and a standard deviation of 3.4. Note that, even though the SLMSP heuristic does not guarantee reaching feasibility, this was the case in every instance under study.

Naturally, the superior performance of the ILMSP in terms of solution quality comes with a cost in terms of longer solution times. While on average the ILMSP takes 535 s to solve an instance, the SLMSP takes only 10.8 s. In other words, the sequential approach is approximately 50 times faster than the integrated one. In summary, these results show that the integrated approach for the LMSP provides incremental benefits in the order of 4% against doing this process sequentially, but at a cost of increasing solution times fifty-fold.

For the sake of completeness, we also implemented a version of the ILMSP where the whole model was fed into the solver without using a callback function. This approach was clearly worse in terms of solution time and quality. For example, in large instances, such as (11/3/415), the solver did not even produce a feasible solution during the time limit imposed.

Fig. 1 gives a closer look into the inner workings of the sequential approach. The dashed bars indicate the percentage of jobs that need to be reassigned to a different maintenance opportunity in the second stage of the heuristic to deal capacity violations from the job assignment. The solid dark bars indicate the variation in the objective function between the two stages of the SLMSP algorithm.

We can observe from this figure that, in every instance except for (3/4/112), the second stage of the SLMSP algorithm needs to reassign jobs to a different maintenance opportunity, which shows that the first stage of the heuristic produces infeasible solutions from an operative perspective. The percentage of jobs that needs to be reassigned ranges between 0% and 11.8%, with an average of 4.1%. The impact of these reassignments, displayed in black bars, reveal that, on average, the objective function value increases by 4.6% when some jobs need to be reassigned in the second stage of the heuristic. Even though this percentage is similar to the number of jobs reallocated, there is little correlation between them. For example, in instance (5/2/94) the SLMSP reallocates 1.1% of its jobs, increasing the objective function in 8%, or in instance (10/3/382) a reassignment of 11.8% of the jobs entail an increase of 7.3% in the objective function. These results show that integrating job scheduling to the maintenance opportunity assignment problem is crucial, since even a small number of job reassignments, may have a substantial impact on the quality of the solution.

5.2.2. Effect of the level of congestion in the system

The congestion level of an instance, i.e., the level of demand (number of jobs and their resource requirements) against the total capacity available is an important factor to take into consideration for designing the maintenance network of an airline. In this section we carry out an experiment to explore the impact of varying the level of system capacity, while keeping fixed level of demand. The level of capacity was modified by applying a factor $\psi \in \{1, 1.2, 1.4, 1.6, 1.8, 2\}$ on the resource availability at each airport, and by closing one of the three available maintenance bases. For this sensitivity analysis, we took instance (11/3/415) as a reference, which is a reasonably challenging instance that should

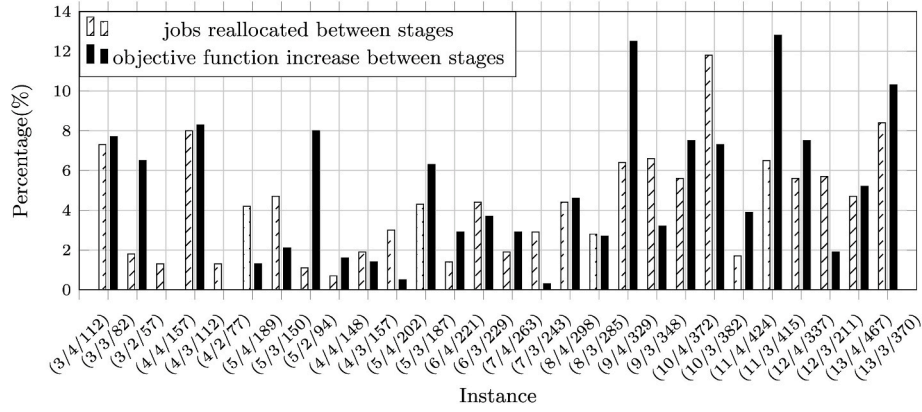


Fig. 1. Performance comparison of the solution stages of the SLMSP algorithm.

capture adequately the different complexities of the problem.

Fig. 2 plots the objective function value for different levels of resource availability ψ , for the case where four original maintenance bases are operative (Fig. 2 (a)) and where only three of these bases are operative (Fig. 2 (b)). Both plots show that the total deviation costs of the schedule (i.e., the objective function, O.F.) decrease as the available resources increase, as expected. By comparing Fig. 2 (a) and (b), we conclude that closing a maintenance base has a significant impact on the rise in the objective value, which goes from 611 (when all maintenance bases are available) to 836, a surge of 36.8%. Moreover, we observe that even if the available resources at the other maintenance bases double, the objective function value in this case (824) is still 34.9% higher than the one obtained when operating all maintenance bases with only half of the resources. This experiment shows that at least in some cases it is preferable to have less total capacity spread in more bases, than more capacity concentrated in fewer bases.

Fig. 3 shows in detail the effect of the number of bases versus capacity illustrated with the example from last paragraph. The three depicted series represent the distribution of job deviation from due time (i.e., earliness or tardiness of the scheduled jobs) in terms of flying hours. The densely dashed gray line with triangle marks corresponds to the scenario with three maintenance bases and $\psi = 1$. The gray solid line with square marks shows the effect on this scenario of doubling capacity, which looks marginal when compared against the effect of opening a fourth maintenance base. We can also observe for these three instances that more than 90% of the jobs are scheduled within six equivalent hours of their due-times. However, when we consider a tighter interval, performance varies visibly: opening the fourth base

allows over a 52% of the jobs to be planned within an hour of their due times, while in both other cases this number is about 40%.

The results of Figs. 2 and 3 show, in summary, that giving spatial flexibility by having distributed maintenance bases across the network can be substantially more convenient than concentrating maintenance operations at fewer airports with higher capacity.

6. Conclusion

As new technologies emerge, aircraft maintenance jobs keep getting more efficient and less time consuming. This opens the possibility of gradually shifting many of these jobs from planned block checks (which force an aircraft to remain on the ground from some hours to days or weeks) to the line maintenance environment, where routine maintenance jobs are performed during layovers, increasing the efficiency of fleet utilisation. However, traditional maintenance planning approaches focus on scheduling heavier jobs, without dealing explicitly with scheduling jobs in line maintenance.

In this paper we introduce the Line Maintenance Scheduling Problem (LMSP). This problem involves two levels of planning. On an aggregate level, the objective is to assign maintenance jobs to available slots or opportunities, aiming to minimise the total deviation from the due time for each job. On an operational level, the objective is to define the exact time within the maintenance opportunity that each job should start, in order to ensure that resource utilisation at each airport or maintenance base is not exceeded at any moment in the planning horizon.

We propose a mathematical formulation for the LMSP that can be used to evaluate and optimise the line maintenance schedule of a fleet of

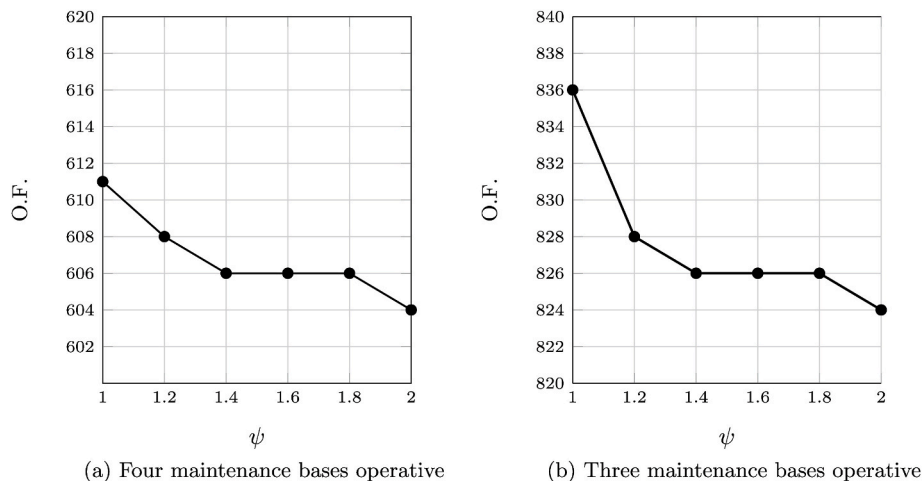


Fig. 2. Total deviation costs (O.F.) versus different levels of supply resources.

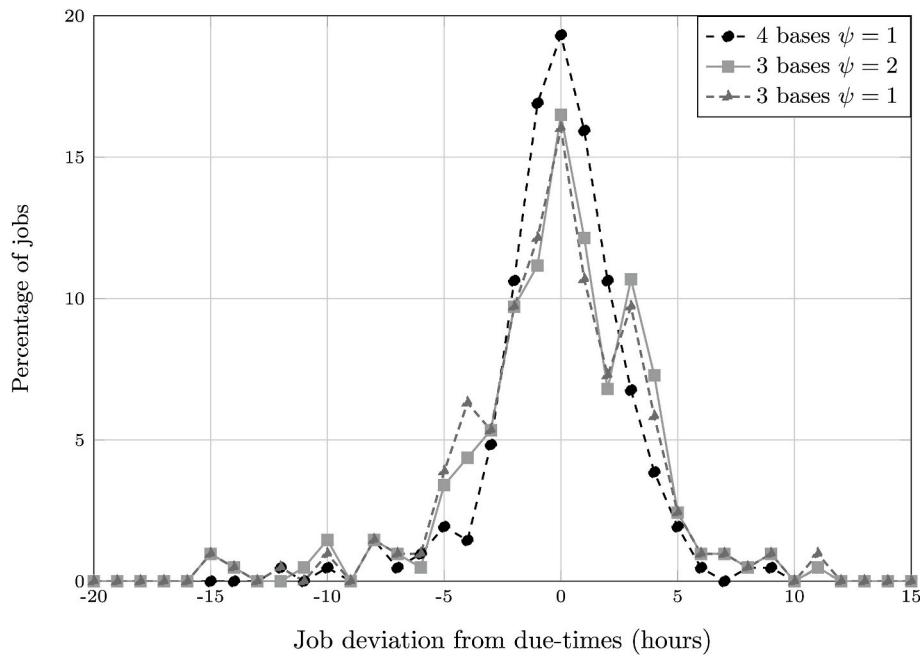


Fig. 3. Job deviation distribution for jobs with deadline in hours.

aircraft. Our model takes the network information (i.e., the list of maintenance opportunities, which can be derived from the aircraft routes, and the resource availability at each airport) and the demand information (a list of jobs to be performed and their resource consumption) and our model finds the feasible schedule that minimises the total deviation of the jobs. Our model can also deal with non-mandatory jobs, which can be postponed for a fixed cost.

Based on this mathematical framework, we designed and tested two different solution approaches for the problem. The first one consisted on an exact algorithm that solved the optimisation model integrating job assignment and timetabling decisions (i.e., the ILMSP algorithm). The second approach consisted of a greedy-type sequential heuristic (the SLMSP algorithm), which first finds a solution for the assignment problem, and then adjusts the timetable to find feasibility at the operational level.

We tested both approaches in a set of 29 instances covering different combinations of number of jobs, number of aircraft, maintenance opportunities, job process times and resource requirements. Our experiments show that the proposed integrated approach, ILMSP, is capable of solving the whole set of instances to optimality on less than 10 min on average. The sequential approach, SLMSP, reached solutions that were 3.5% worse than the obtained by ILMSP, but its solution times averaged nearly 10 s. This suggests that both proposed solution procedures can serve as an operational decision-making tool that can be used routinely to re-optimize the line maintenance schedule whenever new maintenance requirement information becomes available, in a rolling horizon fashion.

Our experiments also revealed that in realistic settings it is convenient to design a maintenance network where capacity is spread spatially, even at expense of the total level of capacity deployed. We provide a concrete example of an instance where opening a fourth maintenance base (i.e., amplifying total capacity in 4/3) led to a considerably better system performance than doubling the capacity at

the three initially available facilities.

Some of the avenues for future research that we envision for this work include the incorporation of more complex relations between jobs in our formulation, such as job precedence constraints, recurring jobs, simultaneity or exclusivity of jobs, and specific arrival or departure time requirements. This would allow our model to better represent a real aircraft line maintenance environment, and make it more useful as a decision-making tool. Another extension to our model would be to extend it to consider the capacity planning aspect of the problem, i.e., to adapt it to answer to the question of how to open or close maintenance bases, or how to allocate capacity on them. Finally, due to the volatility of air traffic operations, a necessary, but also challenging extension to this work would be to incorporate smart ways to deal with uncertainty. Stochastic and/or robust programming implemented in a rolling horizon solution environment, or even a dynamic version of the LMSP to deal with disruptions and emergencies would certainly be a contribution to keep making aircraft maintenance more efficient, and air travel safer and more accessible.

CRedit authorship contribution statement

Syed Shaukat: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing - original draft. **Mathias Katscher:** Methodology, Software, Validation, Data curation. **Cheng-Lung Wu:** Conceptualization, Methodology, Validation, Writing - original draft, Resources, Supervision, Project administration. **Felipe Delgado:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Project administration, Supervision. **Homero Larrain:** Conceptualization, Methodology, Validation, Investigation, Writing - original draft, Writing - review & editing, Project administration, Supervision.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jairtraman.2020.101914>.

The SLMSP heuristic is divided in three main stages: data pre-processing, job assignment and timetabling stages. A complete description of the

heuristic is provided as follows.

Data pre-processing

1. Sort mandatory jobs ($i \in O$) in a non-increasing order of job priority, so jobs with higher priority will be scheduled first. The priority value of each job is assigned according to a priori knowledge of maintenance jobs. For those jobs with the same priority value, we further sort those jobs in a non-decreasing order of job deadlines, so jobs with the same priority value but earlier deadlines will be scheduled first.
2. Sort non-mandatory jobs ($i \in N$) according to the Shortest Processing Time (SPT) principle. To ensure that large jobs can be accommodated during the scheduling process, we insert every n -th position the largest job that has not been inserted yet. The goal is to schedule jobs without deadlines as many as possible when a maintenance opportunity is available, so smaller jobs are scheduled first by SPT. To avoid the inability to schedule large jobs without deadlines, we bring large jobs forward, so these jobs will not be pushed further away from maintenance opportunities. We tried different values of n from 5 to 20. In all cases, the algorithm was able to schedule all non-mandatory jobs, thus not producing any change in the objective function value. For all tested instances, we finally set $n = 5$.
3. For each aircraft, group all the maintenance opportunities belonging to this aircraft across the planning horizon and sort them according to in-house scheduling preferences of the study airline. This preference is driven by a few factors including: (a) preferred maintenance airports for certain types of jobs; (b) the availability of person-hours for specific types of jobs at different airports; and (c) the availability of spare parts and specialised tools for maintenance jobs.
4. For each job i , we define q_i as the variable job release time. We set q_i such that $Q_i \leq q_i \leq C_i$. The idea is to schedule each job closer to its due-time C_i as possible.

Job assignment stage

This process tries to assign jobs to maintenance opportunity, going through the ordered set J starting from the first element that corresponds to the job with the highest priority. For each job $i \in O$ the assignment process is summarised in [Algorithm 1](#).

Algorithm 1. (Job assignment process for mandatory jobs)

```

1: Initialize ( $scheduling, q_i, z_i^*, y_i^*$ )  $\leftarrow$  ( $True, C_i, \infty, 0$ ).
2: while  $scheduling$  do
3:   Generate a set  $M_i$  of maintenance opportunities where it is feasible to schedule  $i$ .
4:   if  $M_i \neq \emptyset$  then
5:      $scheduling \leftarrow False$ 
6:     for  $m \in M_i$  do
7:       Compute the tardiness  $z_i$  and earliness  $y_i$ .
8:       if  $\max(z_i^*, y_i^*) > \max(z_i, y_i)$  then
9:          $(z_i^*, y_i^*, m_i) \leftarrow (z_i, y_i, m)$ 
10:      end if
11:    end for
12:  else if  $q_i \geq Q_i$  then
13:    Update the variable release time:  $q_i \leftarrow q_i - \frac{D_i - Q_i}{10}$ 
14:  else
15:    Mark job  $i$  as unschedule and  $scheduling \leftarrow False$ 
16:  end if
17: end while
18: Update the remaining resource  $k$  availability at maintenance opportunity  $m_i$  such that  $R_{m_i}^k \leftarrow R_{m_i}^k - r_i^k, \forall k \in K$ .

```

The set M_i is composed by all maintenance opportunities that belong to an specific aircraft that meets the following scheduling requirements: i) the process time of job i is within the limit of m such that $T_i \leq u_m - l_m$; ii) the allocation of job i to m is within the desired earliness or tardiness such that $t_m \in [q_i, D_i], \forall n \in S$; and iii) the amount of resource k , r_i^k required to carry out job i , is within the resource capacity, R_p^k available at airport p , $\forall k \in K$.

For jobs without deadline (non-mandatory), the allocation process is simpler than the one shown in [Algorithm 1](#). If the set $M_i \neq \emptyset$ then assign job i to any of the maintenance opportunity in M_i and update the available resources. In other case, mark the job as unscheduled.

Timetabling stage

This stage begins by setting all jobs to start at the beginning of the maintenance opportunity they were assigned during the job assignment stage. For each maintenance base p , the operational planning process is depicted in [Algorithm 2](#). This algorithm does not guarantee finding a feasible solution.

Algorithm 2. (Timetabling stage for mandatory jobs)

```

1: Initialize infeasible  $\leftarrow$  True.
2: while infeasible do
3:   infeasible  $\leftarrow$  False
4:   Generate the demand profile of each resource k for the entire planning horizon.
5:   Starting from the beginning of the planning horizon, check if demand exceeds capacity for any
     resource type k.
6:   if demand > capacity then
7:     infeasible  $\leftarrow$  True
8:     Generate set G of conflicting jobs.
9:     Generate set G1  $\subseteq$  G composed of jobs i  $\in$  G whose removal would restate feasibility of the
       detected violation.
10:    if G1  $\neq$   $\emptyset$  then
11:      Choose job i  $\in$  G1 that demands the less resources of the critical type.
12:      Set the starting time of job i as  $s_i = \min_{j \in G} (e_j)$ 
13:    else
14:      Select from the set G a certain job l that in case of being reassigned to another maintenance
        opportunity ml minimises its deviation from due time.
15:      Assign job l to maintenance opportunity ml
16:    end if
17:  end if
18:  Update profile
19: end while

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